

Predictors of Absolute VO2max

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Setup

Combine Raw Data Files

This step was run once to combine the individual Excel files into `combined_tests.csv` and is not re-run after.

Data Cleaning

```
df_raw <- read.csv(here("data", "combined_tests.csv"))

df_raw <- df_raw |>
  rename(file = "source_file")

df_raw_sum <- read.csv(here("data", "Data_Summary_with_age.csv"))

df_raw_sum <- df_raw_sum |>
  select("file", "age")

df_raw <- df_raw |>
  group_by(file) |>
  mutate(max_bla = max(bla))

### Making a df that only has variables that are needed for the study

colnames(df_raw)

df_clean <- df_raw |>
  select("file", "gender", "event", "test", "height", "weight", "bmi", "fat", "lbm", "hb",
        "hct", "rbc", "wbc", "max_bla", "vo2max", "vo2max_absolute", "hrmax", "load_tlac",
        "hr_tlac", "percent_vo2max_tlac", "percent_h_rmax_tlac", "bla_tlac")

df_clean <- merge(df_clean, df_raw_sum, by = "file")

df_clean <- df_clean |>
  relocate("age", .after = gender)

df_clean <- df_clean |>
  distinct(file, .keep_all = T)

df_clean <- df_clean |>
  filter(!file %in% c("A539.xlsx", "A140.xlsx", "A562.xlsx", "A1065.xlsx"))

### Df manipulation and cleaning for analysis
colnames(df_clean)

anyNA(df_clean)

colSums(is.na(df_clean))

df_clean$weight[df_clean$weight == "Missing"] <- NA

df_clean <- df_clean |>
  na.omit() |>
  mutate(weight = as.numeric(as.character(weight)))

df_clean <- df_clean |>
  mutate(across(where(is.numeric), ~ round(., 2)))

df_clean <- df_clean |>
```

```

filter(age >= 18)

df_clean <- df_clean |>
  rename(sex = gender)

df_clean <- df_clean %>%
  mutate(test = case_match(test,
    c("Rowing", "Kayak") ~ "watersports",
    .default = test
  ))

```

Cleaning notes

- Four participants (A539, A140, A562, A1065) were excluded because of implausible or irreconcilable values (VO2max, body fat %, hematocrit).
- Participants under 18 or with missing values were removed, and rowing and kayak were combined into “watersports”.
- Blood lactate(bla) used max() function due to bla being linked to max effort in GxT
- Final analytic sample: **N = 699** (147 women, 552 men).

Descriptive Statistics

```

# Overall
df_clean |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute)
  ) |>
  knitr::kable(digits = 2, caption = "Overall sample")

```

Table 1: Overall sample

n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
699	27.56	7	4.61	0.86

```

# By modality
df_clean |>
  group_by(test) |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute),
  )

```

```
.groups = "drop"
) |>
knitr::kable(digits = 2, caption = "By modality")
```

Table 2: By modality

test	n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
Cycling	247	30.38	6.93	4.58	0.74
Running	225	29.37	6.89	4.26	0.88
watersports	227	22.68	3.98	4.99	0.81

```
# By sex
df_clean |>
  group_by(sex) |>
  summarise(
    n = n(),
    age_mean = mean(age),
    age_sd = sd(age),
    abs_vo2_mean = mean(vo2max_absolute),
    abs_vo2_sd = sd(vo2max_absolute),
    .groups = "drop"
  ) |>
knitr::kable(digits = 2, caption = "By sex")
```

Table 3: By sex

sex	n	age_mean	age_sd	abs_vo2_mean	abs_vo2_sd
F	147	28.24	6.07	3.40	0.49
M	552	27.37	7.22	4.93	0.62

```
# Check that there are enough participants in each moderator group
df_clean |>
  count(sex) |>
  mutate(percent = round(n / sum(n) * 100, 2)) |>
  knitr::kable(caption = "Sex distribution")
```

Table 4: Sex distribution

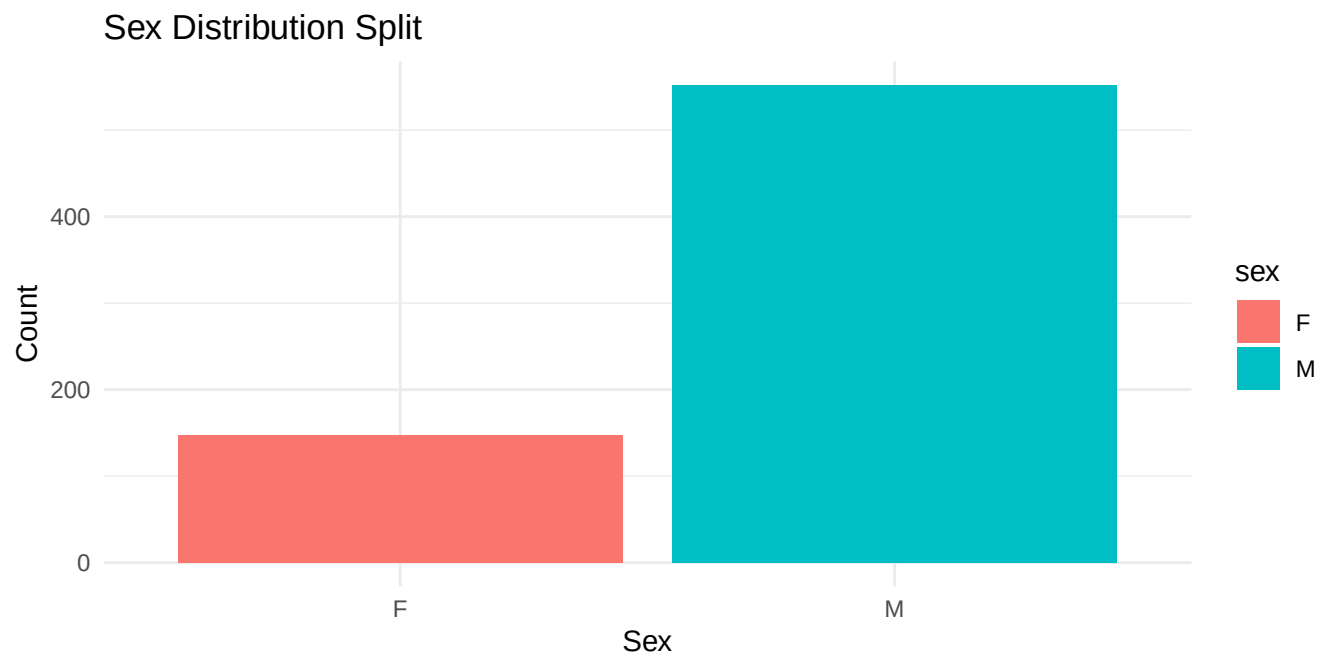
sex	n	percent
F	147	21.03
M	552	78.97

```
df_clean |>
  count(test) |>
  mutate(percent = round(n / sum(n) * 100, 2)) |>
  knitr::kable(caption = "Modality distribution")
```

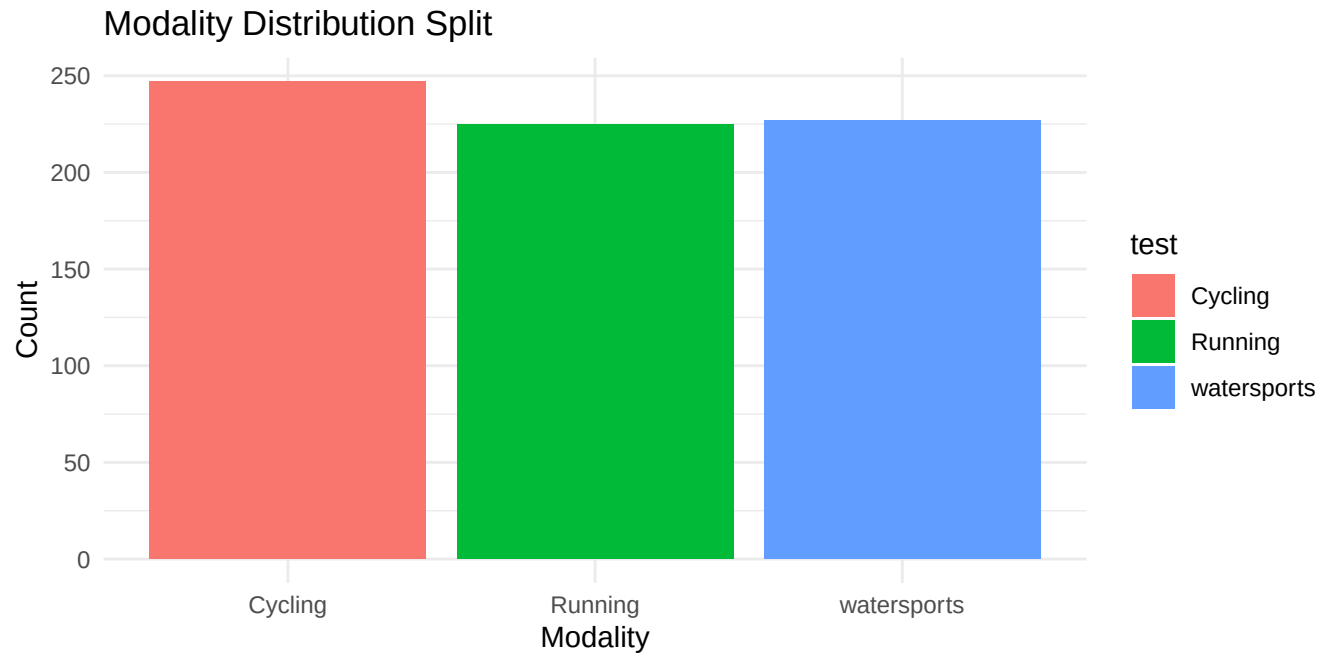
Table 5: Modality distribution

test	n	percent
Cycling	247	35.34
Running	225	32.19
watersports	227	32.47

```
ggplot(df_clean, aes(x = sex, fill = sex)) +
  geom_bar() +
  labs(title = "Sex Distribution Split", y = "Count", x = "Sex") +
  theme_minimal()
```



```
ggplot(df_clean, aes(x = test, fill = test)) +
  geom_bar() +
  labs(title = "Modality Distribution Split", y = "Count", x = "Modality") +
  theme_minimal()
```



Research Question 1

How well do the known predictors (body fat %, lean body mass, hemoglobin, hematocrit) jointly explain absolute VO2max, without moderators? Body weight is left out because it is almost fully determined by fat % and lean mass.

Model and coefficients

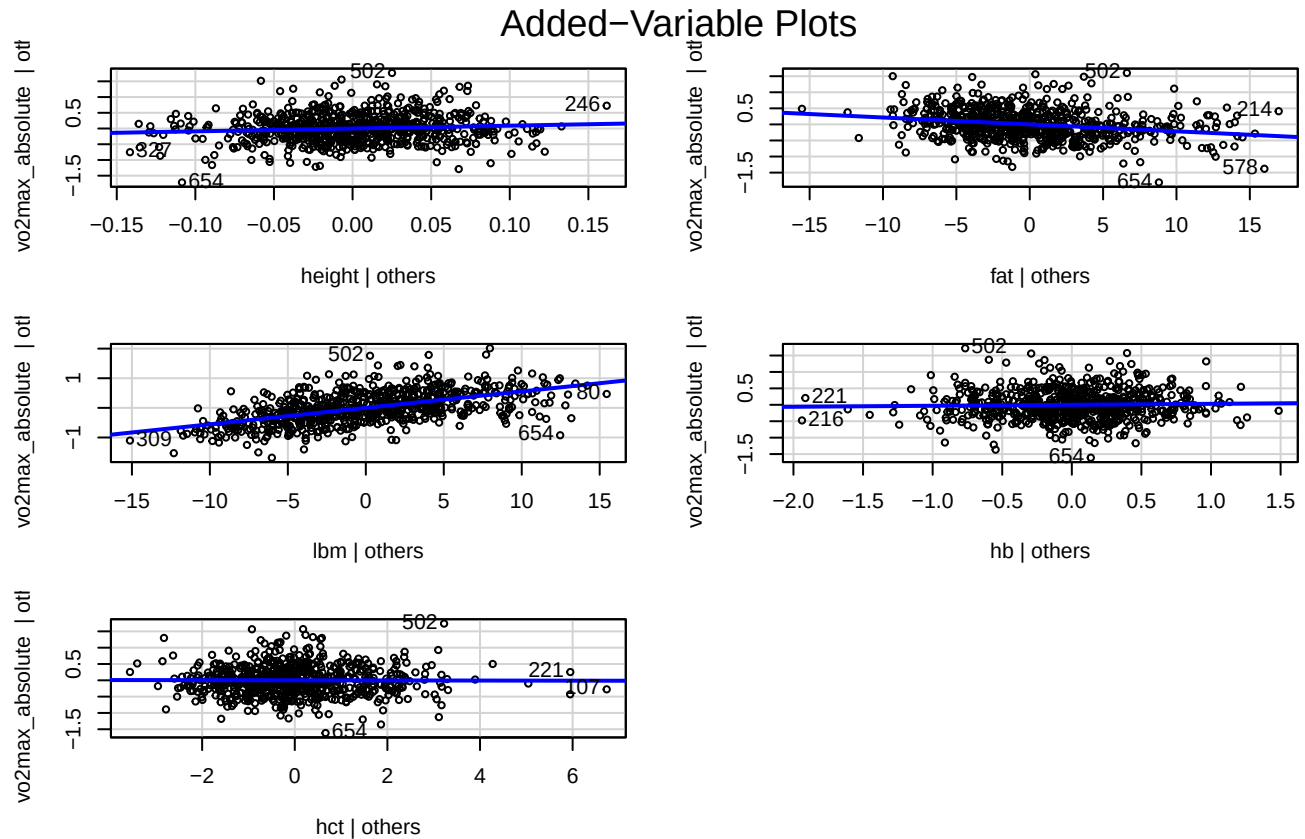
```
A1 <- lm(vo2max_absolute ~ height + fat + lbm + hb + hct, data = df_clean)
```

```
summary(A1)
```

```
##
## Call:
## lm(formula = vo2max_absolute ~ height + fat + lbm + hb + hct,
##     data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.6156 -0.2798 -0.0298  0.2654  1.7436
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.531663   0.580964  -0.915   0.3604
## height       0.902733   0.360568   2.504   0.0125 *
## fat         -0.021519   0.003448  -6.241 7.59e-10 ***
## lbm          0.055412   0.003234  17.134 < 2e-16 ***
## hb           0.030255   0.036468   0.830   0.4070
```

```
## hct          -0.001897   0.012931  -0.147   0.8834
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4534 on 693 degrees of freedom
## Multiple R-squared:  0.7243, Adjusted R-squared:  0.7223
## F-statistic: 364.2 on 5 and 693 DF,  p-value: < 2.2e-16
```

```
avPlots(A1)
```



```
tidy(A1, conf.int = TRUE) |> tab("RQ1: Regression coefficients")
```

Table 6: RQ1: Regression coefficients

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	-0.532	0.581	-0.915	0.360	-1.672	0.609
height	0.903	0.361	2.504	0.013	0.195	1.611
fat	-0.022	0.003	-6.241	<0.001	-0.028	-0.015
lbm	0.055	0.003	17.134	<0.001	0.049	0.062
hb	0.030	0.036	0.830	0.407	-0.041	0.102
hct	-0.002	0.013	-0.147	0.883	-0.027	0.023

```
glance(A1) |>
  select(r.squared, adj.r.squared, sigma, statistic, p.value, nobs) |>
  tab("RQ1: Overall model fit")
```

Table 7: RQ1: Overall model fit

r.squared	adj.r.squared	sigma	statistic	p.value	nobs
0.724	0.722	0.453	364.165	<0.001	699

Interpretation: Together the four predictors explain about 72% of the variance in absolute VO2max ($R^2 = 0.72$). Lean mass is the strongest predictor: each additional kg is associated with about 0.061 L/min more VO2max (95% CI 0.057 to 0.065). Each additional percentage point of body fat is associated with about 0.023 L/min less (CI -0.030 to -0.016). Hemoglobin and hematocrit are not associated with VO2max once body composition is accounted for (their CIs include zero).

Multicollinearity

```
vif(A1) |>
  enframe(name = "predictor", value = "VIF") |>
  knitr::kable(digits = 2, caption = "RQ1: Variance inflation factors")
```

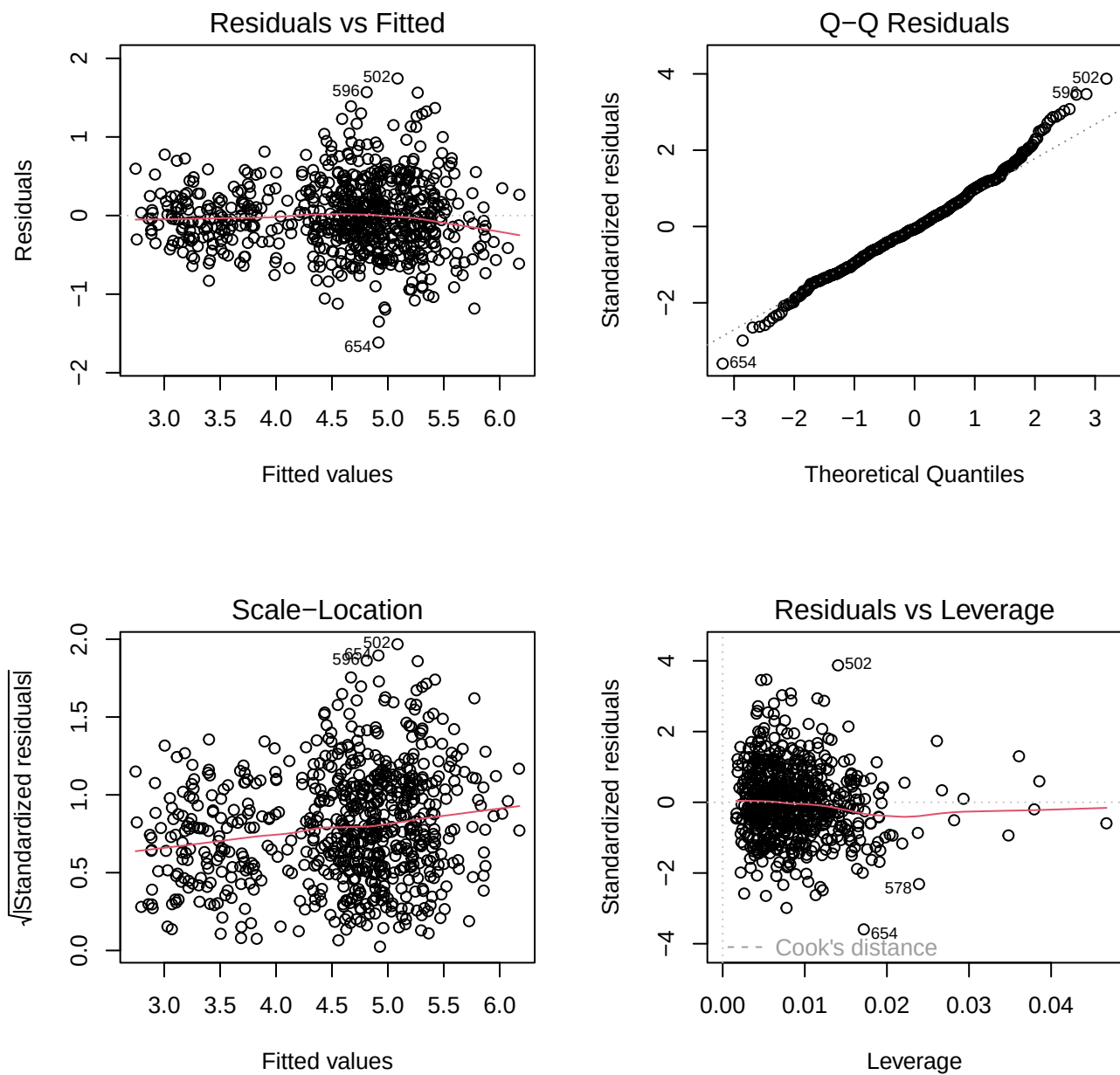
Table 8: RQ1: Variance inflation factors

predictor	VIF
height	3.41
fat	1.42
lbm	3.83
hb	4.88
hct	4.69

Interpretation: Fat % (1.4) and lean mass (1.6) show little collinearity. Hemoglobin (4.9) and hematocrit (4.7) are correlated with each other, as expected, so their individual coefficients are less precise. This does not affect the joint test of the blood variables below.

Assumption checks

```
par(mfrow = c(2, 2)); plot(A1); par(mfrow = c(1, 1))
```



```
bptest(A1) |> tidy() |> tab("Breusch-Pagan test for heteroscedasticity")
```

Table 9: Breusch-Pagan test for heteroscedasticity

statistic	p.value	parameter	method
23.381	<0.001	5	studentized Breusch-Pagan test

Interpretation: The residuals are centered around zero with no curvature, and the Q-Q plot follows the line. No observation has a large Cook's distance. The Breusch-Pagan test detects some non-constant variance

($p < 0.001$), so robust standard errors are used next as a check.

Do the blood variables add anything?

```
A1_body <- lm(vo2max_absolute ~ fat + lbm + height, data = df_clean)

anova(A1_body, A1) |>
  tidy() |>
  mutate(model = c("fat + lbm + height", "fat + lbm + height + hb + hct + max_bla"), .before = 1) |>
  select(-term) |>
  tab("Model comparison: body composition only vs. adding hb and hct")
```

Table 10: Model comparison: body composition only vs. adding hb and hct

model	df.residual	rss	df	sumsq	statistic	p.value
fat + lbm + height	695	142.852	NA	NA	NA	NA
fat + lbm + height + hb + hct + max_bla	693	142.486	2	0.366	0.89	0.411

Interpretation: Adding hemoglobin and hematocrit to a model with fat % and lean mass does not improve fit ($F = 0.92$, $p = 0.40$). Body composition accounts for the explainable variance, and the blood variables add essentially nothing.

Research Question 2

Do age, sex, or exercise modality moderate the associations between the known predictors and absolute VO2max?

Models

```
df_clean <- df_clean |>
  mutate(sex = factor(sex), test = factor(test))

# For each moderator: a main-effects model and a full model with all predictor x moderator interactions
A2_age_main <- lm(vo2max_absolute ~ height + fat + lbm + hb + hct + age, data = df_clean)
A2_age      <- lm(vo2max_absolute ~ (height + fat + lbm + hb + hct) * age, data = df_clean)

summary(A2_age)

##
## Call:
## lm(formula = vo2max_absolute ~ (height + fat + lbm + hb + hct) *
##     age, data = df_clean)
##
## Residuals:
```

```

##      Min      1Q   Median      3Q      Max
## -1.49758 -0.28390 -0.01524  0.26866  1.73430
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  1.5968360  2.4547998   0.650  0.5156
## height      -2.7321217  1.4776116  -1.849  0.0649 .
## fat          0.0144810  0.0150118   0.965  0.3351
## lbm          0.0869972  0.0133112   6.536 1.23e-10 ***
## hb           0.2003086  0.1381979   1.449  0.1477
## hct          -0.0111527  0.0494455  -0.226  0.8216
## age          -0.0601338  0.0869014  -0.692  0.4892
## height:age    0.1180956  0.0523989   2.254  0.0245 *
## fat:age       -0.0013012  0.0005287  -2.461  0.0141 *
## lbm:age       -0.0010584  0.0004658  -2.272  0.0234 *
## hb:age        -0.0061998  0.0047992  -1.292  0.1968
## hct:age        0.0004018  0.0017183   0.234  0.8152
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4447 on 687 degrees of freedom
## Multiple R-squared:  0.7371, Adjusted R-squared:  0.7329
## F-statistic: 175.1 on 11 and 687 DF,  p-value: < 2.2e-16

A2_sex_main <- lm(vo2max_absolute ~ height + fat + lbm + hb + hct + sex, data = df_clean)
A2_sex      <- lm(vo2max_absolute ~ (height + fat + lbm + hb + hct) * sex, data = df_clean)

summary(A2_sex)

##
## Call:
## lm(formula = vo2max_absolute ~ (height + fat + lbm + hb + hct) *
##     sex, data = df_clean)
##
## Residuals:
##      Min      1Q   Median      3Q      Max
## -1.5599 -0.2643 -0.0326  0.2539  1.7614
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.561339  1.457437  -0.385  0.7002
## height       0.509771  0.957297   0.533  0.5945
## fat          -0.002153  0.006819  -0.316  0.7523
## lbm           0.059496  0.010737   5.541 4.29e-08 ***
## hb           0.137463  0.079045   1.739  0.0825 .
## hct          -0.037834  0.026958  -1.403  0.1609
## sexM          0.772912  1.600379   0.483  0.6293
## height:sexM  0.449653  1.035315   0.434  0.6642
## fat:sexM     -0.020434  0.008414  -2.429  0.0154 *
## lbm:sexM     -0.009829  0.011450  -0.858  0.3909
## hb:sexM      -0.149392  0.089251  -1.674  0.0946 .
## hct:sexM      0.040597  0.030694   1.323  0.1864
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

```

```
##
## Residual standard error: 0.4499 on 687 degrees of freedom
## Multiple R-squared:  0.731, Adjusted R-squared:  0.7267
## F-statistic: 169.7 on 11 and 687 DF,  p-value: < 2.2e-16

A2_test_main <- lm(vo2max_absolute ~ height + fat + lbm + hb + hct + test, data = df_clean)
A2_test      <- lm(vo2max_absolute ~ (height + fat + lbm + hb + hct) * test, data = df_clean)

summary(A2_test)
```

```
##
## Call:
## lm(formula = vo2max_absolute ~ (height + fat + lbm + hb + hct) *
##     test, data = df_clean)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.95189 -0.27200 -0.02149  0.27887  1.64597
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)   -0.048197   1.039052  -0.046  0.96302
## height         1.809030   0.635727   2.846  0.00457 **
## fat           -0.038456   0.005972  -6.439 2.27e-10 ***
## lbm            0.044834   0.005116   8.763 < 2e-16 ***
## hb            -0.026308   0.066601  -0.395  0.69296
## hct           -0.009246   0.023649  -0.391  0.69594
## testRunning    0.350699   1.453181   0.241  0.80937
## testwatersports -1.085557   1.531217  -0.709  0.47860
## height:testRunning -1.629042   0.892140  -1.826  0.06829 .
## height:testwatersports -1.717061   0.919780  -1.867  0.06236 .
## fat:testRunning   0.013833   0.008977   1.541  0.12383
## fat:testwatersports 0.036027   0.008670   4.155 3.66e-05 ***
## lbm:testRunning   0.015428   0.007932   1.945  0.05219 .
## lbm:testwatersports 0.022247   0.008201   2.713  0.00684 **
## hb:testRunning    0.055420   0.090920   0.610  0.54237
## hb:testwatersports 0.156693   0.090334   1.735  0.08327 .
## hct:testRunning   0.011659   0.032553   0.358  0.72034
## hct:testwatersports -0.003634   0.031851  -0.114  0.90920
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4467 on 681 degrees of freedom
## Multiple R-squared:  0.7371, Adjusted R-squared:  0.7305
## F-statistic: 112.3 on 17 and 681 DF,  p-value: < 2.2e-16
```

```
# Model fit comparison
bind_rows(
  "RQ1 (no moderator)" = glance(A1),
  "Age"                 = glance(A2_age),
  "Sex"                 = glance(A2_sex),
  "Modality"            = glance(A2_test),
  .id = "model"
```

```
) |>
  select(model, r.squared, adj.r.squared, sigma) |>
  tab("R-squared with and without moderators")
```

Table 11: R-squared with and without moderators

model	r.squared	adj.r.squared	sigma
RQ1 (no moderator)	0.724	0.722	0.453
Age	0.737	0.733	0.445
Sex	0.731	0.727	0.450
Modality	0.737	0.731	0.447

```
tidy(A2_age) |> tab("Full model: predictors x age")
```

Table 12: Full model: predictors x age

term	estimate	std.error	statistic	p.value
(Intercept)	1.597	2.455	0.650	0.516
height	-2.732	1.478	-1.849	0.065
fat	0.014	0.015	0.965	0.335
lbm	0.087	0.013	6.536	<0.001
hb	0.200	0.138	1.449	0.148
hct	-0.011	0.049	-0.226	0.822
age	-0.060	0.087	-0.692	0.489
height:age	0.118	0.052	2.254	0.025
fat:age	-0.001	0.001	-2.461	0.014
lbm:age	-0.001	0.000	-2.272	0.023
hb:age	-0.006	0.005	-1.292	0.197
hct:age	0.000	0.002	0.234	0.815

```
tidy(A2_sex) |> tab("Full model: predictors x sex")
```

Table 13: Full model: predictors x sex

term	estimate	std.error	statistic	p.value
(Intercept)	-0.561	1.457	-0.385	0.700
height	0.510	0.957	0.533	0.595
fat	-0.002	0.007	-0.316	0.752
lbm	0.059	0.011	5.541	<0.001
hb	0.137	0.079	1.739	0.082
hct	-0.038	0.027	-1.403	0.161
sexM	0.773	1.600	0.483	0.629
height:sexM	0.450	1.035	0.434	0.664
fat:sexM	-0.020	0.008	-2.429	0.015
lbm:sexM	-0.010	0.011	-0.858	0.391
hb:sexM	-0.149	0.089	-1.674	0.095
hct:sexM	0.041	0.031	1.323	0.186

```
tidy(A2_test) |> tab("Full model: predictors x modality")
```

Table 14: Full model: predictors x modality

term	estimate	std.error	statistic	p.value
(Intercept)	-0.048	1.039	-0.046	0.963
height	1.809	0.636	2.846	0.005
fat	-0.038	0.006	-6.439	<0.001
lbm	0.045	0.005	8.763	<0.001
hb	-0.026	0.067	-0.395	0.693
hct	-0.009	0.024	-0.391	0.696
testRunning	0.351	1.453	0.241	0.809
testwatersports	-1.086	1.531	-0.709	0.479
height:testRunning	-1.629	0.892	-1.826	0.068
height:testwatersports	-1.717	0.920	-1.867	0.062
fat:testRunning	0.014	0.009	1.541	0.124
fat:testwatersports	0.036	0.009	4.155	<0.001
lbm:testRunning	0.015	0.008	1.945	0.052
lbm:testwatersports	0.022	0.008	2.713	0.007
hb:testRunning	0.055	0.091	0.610	0.542
hb:testwatersports	0.157	0.090	1.735	0.083
hct:testRunning	0.012	0.033	0.358	0.720
hct:testwatersports	-0.004	0.032	-0.114	0.909

Interpretation: Adding the moderator terms raises R^2 only slightly, from 0.722 in RQ1 to 0.735 (age), 0.730 (sex) and 0.735 (modality). The predictors are not centered, so the main-effect rows (and the sex and modality rows) describe the reference case where the other predictors equal zero and should not be interpreted on their own. The rows to read are the interaction terms (e.g. `fat:age`), which show how a predictor's slope differs by age, sex or modality. The tests below make that clearer.

Step 1: Does each moderator change the relationships at all?

Joint F-test of all four predictor x moderator interactions, with Holm adjustment across the three moderators.

```
bind_rows(
  Age      = tidy(anova(A2_age_main, A2_age)),
  Sex      = tidy(anova(A2_sex_main, A2_sex)),
  Modality = tidy(anova(A2_test_main, A2_test)),
  .id = "moderator"
) |>
  filter(!is.na(statistic)) |>
  select(moderator, df, F_value = statistic, p.value) |>
  mutate(p_holm = p.adjust(p.value, method = "holm")) |>
  tab("Joint test of all interactions per moderator")
```

Table 15: Joint test of all interactions per moderator

moderator	df	F_value	p.value	p_holm
Age	5	2.955	0.012	0.024

moderator	df	F_value	p.value	p_holm
Sex	5	1.942	0.085	0.085
Modality	10	3.109	<0.001	0.002

Interpretation: All three moderators change the predictor relationships, but with very different strength. Modality is clearly significant ($F = 3.79$, Holm-adjusted $p = 0.0007$). Age ($p = 0.044$) and sex ($p = 0.040$) are only just significant after adjustment, so those results are suggestive rather than strong.

Step 2: Which predictors are moderated?

```
Anova(A2_age, type = 2) |> tidy() |> tab("Type II tests: predictors x age")
```

Table 16: Type II tests: predictors x age

term	sumsq	df	statistic	p.value
height	0.347	1	1.756	0.186
fat	7.958	1	40.235	<0.001
lbm	63.508	1	321.106	<0.001
hb	0.120	1	0.606	0.437
hct	0.000	1	0.000	0.999
age	3.691	1	18.665	<0.001
height:age	1.005	1	5.080	0.025
fat:age	1.198	1	6.057	0.014
lbm:age	1.021	1	5.162	0.023
hb:age	0.330	1	1.669	0.197
hct:age	0.011	1	0.055	0.815
Residuals	135.873	687	NA	NA

```
Anova(A2_sex, type = 2) |> tidy() |> tab("Type II tests: predictors x sex")
```

Table 17: Type II tests: predictors x sex

term	sumsq	df	statistic	p.value
height	1.218	1	6.016	0.014
fat	3.076	1	15.202	<0.001
lbm	37.652	1	186.050	<0.001
hb	0.062	1	0.305	0.581
hct	0.052	1	0.256	0.613
sex	1.490	1	7.362	0.007
height:sex	0.038	1	0.189	0.664
fat:sex	1.194	1	5.898	0.015
lbm:sex	0.149	1	0.737	0.391
hb:sex	0.567	1	2.802	0.095
hct:sex	0.354	1	1.749	0.186
Residuals	139.032	687	NA	NA

```
Anova(A2_test, type = 2) |> tidy() |> tab("Type II tests: predictors x modality")
```

Table 18: Type II tests: predictors x modality

term	sumsq	df	statistic	p.value
height	0.724	1	3.629	0.057
fat	7.517	1	37.671	<0.001
lbm	55.267	1	276.952	<0.001
hb	0.356	1	1.782	0.182
hct	0.053	1	0.267	0.605
test	0.386	2	0.968	0.380
height:test	0.912	2	2.285	0.103
fat:test	3.479	2	8.717	<0.001
lbm:test	1.642	2	4.114	0.017
hb:test	0.631	2	1.580	0.207
hct:test	0.052	2	0.131	0.877
Residuals	135.897	681	NA	NA

```
# Direction and size of the age effect from the main-effects model
tidy(A2_age_main, conf.int = TRUE) |>
  filter(term == "age") |>
  tab("Age effect on VO2max (L/min per year), main-effects model")
```

Table 19: Age effect on VO2max (L/min per year), main-effects model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
age	-0.011	0.003	-4.29	<0.001	-0.016	-0.006

Interpretation: The moderation is concentrated in one predictor: body fat. The fat slope differs by age ($p = 0.007$), by sex ($p = 0.005$) and strongly by modality ($p < 0.0001$). The lean mass and blood-variable interactions are not significant (lean mass x modality $p = 0.068$ and hemoglobin x modality $p = 0.086$ are trends only). Age also has an independent main effect on VO2max ($p < 0.001$); the last table gives its direction and size per year.

Step 3: Body fat slope within each group

```
emtrends(A2_test, ~ test, var = "fat") |> as.data.frame() |> tab("Fat slope by modality")
```

Table 20: Fat slope by modality

test	fat.trend	SE	df	lower.CL	upper.CL
Cycling	-0.038	0.006	681	-0.050	-0.027
Running	-0.025	0.007	681	-0.038	-0.011
watersports	-0.002	0.006	681	-0.015	0.010

```
emtrends(A2_sex, ~ sex, var = "fat") |> as.data.frame() |> tab("Fat slope by sex")
```

Table 21: Fat slope by sex

sex	fat.trend	SE	df	lower.CL	upper.CL
F	-0.002	0.007	687	-0.016	0.011
M	-0.023	0.005	687	-0.032	-0.013

```
emtrends(A2_age, ~ age, var = "fat", at = list(age = c(20, 30, 40))) |>
  as.data.frame() |> tab("Fat slope by age")
```

Table 22: Fat slope by age

age	fat.trend	SE	df	lower.CL	upper.CL
20	-0.012	0.005	687	-0.022	-0.001
30	-0.025	0.004	687	-0.032	-0.017
40	-0.038	0.007	687	-0.052	-0.023

Interpretation: These are the effects of one additional percentage point of body fat on VO2max (L/min) within each group. Body fat lowers VO2max most in cyclists (about -0.04) and less in runners (about -0.03), while in watersports athletes the slope is close to zero. Among men the fat slope is about -0.026, while among women it is close to zero. The fat penalty also grows with age (more negative at 40 than at 20). Check the confidence intervals: slopes whose CIs include zero mean body fat is not clearly related to VO2max in that group.

Summary

- Lean mass (positive) and body fat % (negative) explain about 72% of the variance in absolute VO2max. Hemoglobin and hematocrit add nothing beyond them.
- Age, sex and modality moderate these relationships mainly through body fat. The modality effect is the strongest, and the age and sex effects are more modest.
- Age, sex and modality differ from each other in this sample (watersports athletes are younger and mostly male), so the three fat interactions may partly reflect the same underlying difference.